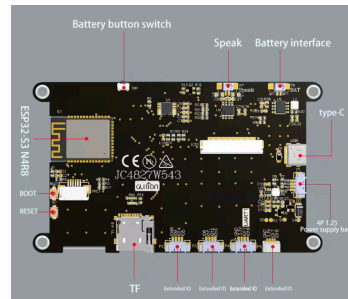


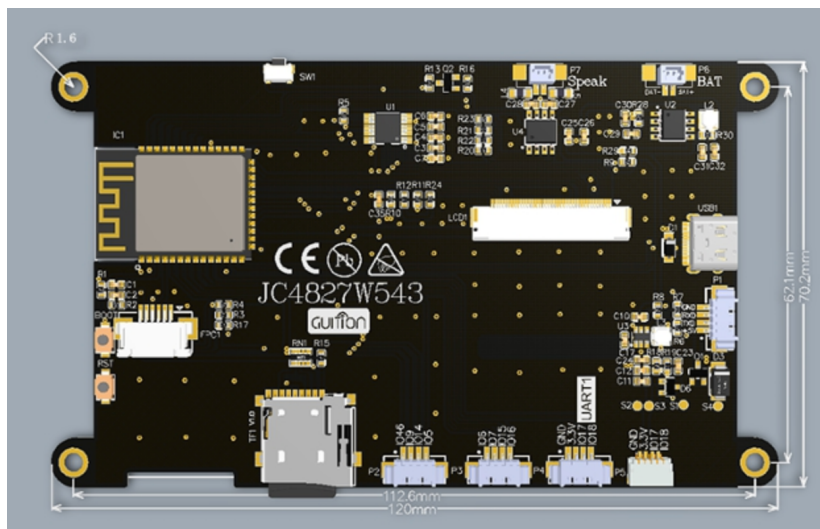
Display Casing

Screen Dimensions:



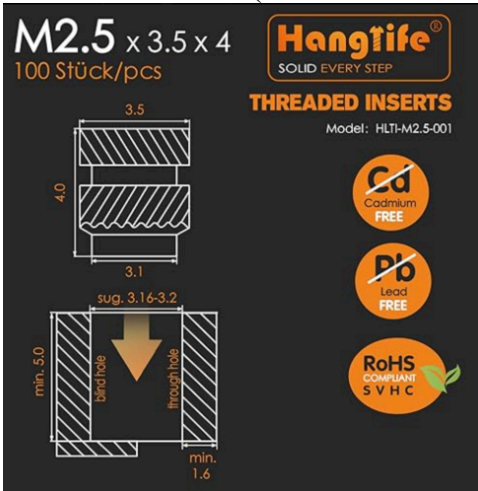
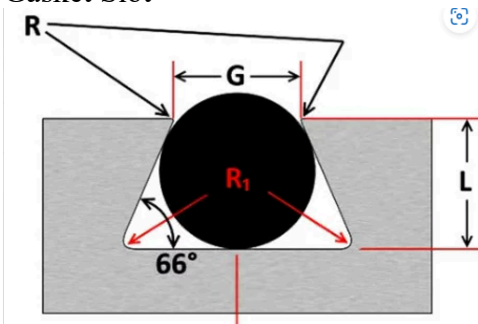
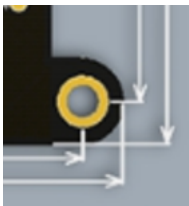
Hardware Feature of JC4827W543C/R_I

4.3" LCD-TFT HMI Capacitive touch or Resistance touch display module
Supports WiFi and Bluetooth
Support lithium battery power supply
Capacitive touch or Resistance touch
480 x 272 screen resolution
RGB 65K true-to-life colours
Onboard 240MHz LX6 MCU
520K Byte RAM
8 MB PSRAM
4 MB Flash memory
JST1.25 4 Pins (+5V, TX, RX, GND) TTL serial interface



Acrylic Cutout: 95x53.50x3mm

Detail Dimensions:

Part	Exact	Measurement with Tolerance
Pegs	3.2mm	2.5mm
Minimum height of screen	5mm	
Height of screen + wire allowance	15mm	
Heat Screw inserts (on bottom side of case)	3.5mm	3.2mm cylindrical holes 3.6mm hole taper in (0.2mm, 45 degrees chamfer)
		
Screw holes	2M	4mm diameter head insert
Gasket Slot	1.5mm (diameter)	1.3mm (diameter) = G = L
		
Diameter of round stick outs	7.4mm (diameter)	7.9mm (diameter)
		

- Gasket only on bottom side of case
- The gasket should be interior to the screws, meaning the screws should be placed outside the gasket's sealing area.
 - **Potential Leakage** – If the screws are within the gasket area, water and dust could bypass the gasket through the screw holes.

- **Compression Concerns** – The screws should clamp the enclosure **outside** the gasket to ensure even pressure distribution. Right now, the gasket may not be compressed effectively.
- **Risk of Gasket Failure** – Placing screws inside the gasket could create stress points, leading to uneven compression or long-term wear on the seal.
- Gasket slot geometry:
 - Distance between screen and acrylic = 3.5mm
 - Cable gland diameter = 12mm

Sourcing Parts

Part	Source			
M2 Screw 10mm (x14 screws)	https://www.engsoc.uwaterloo.ca/resources/ridgidware/ <table><tr><td>4900225.4</td><td>M2 10MM SOCKET CAP SCRE</td><td>0.10</td></tr></table>	4900225.4	M2 10MM SOCKET CAP SCRE	0.10
4900225.4	M2 10MM SOCKET CAP SCRE	0.10		
M4 Screw 10mm (x2 screws)	<table><tr><td>4900314.4</td><td>M4 10MM SOCKET CAP SCRE</td><td>0.20</td></tr></table>	4900314.4	M4 10MM SOCKET CAP SCRE	0.20
4900314.4	M4 10MM SOCKET CAP SCRE	0.20		
Heat inserts for screws (for M2)	https://www.amazon.ca/HANGLIFE-Heat-Set-Threaded-Printing-Components/dp/B0CS6XJSSL/ref=sr_1_4?crd=2RBG6TT49DFNV&dib=eyJ2IjojMSJ9.dsqn0FN_Goi2Hf_w6WWRI2ZVGHeDNyF2WxO9y4oMRY-T8G13MZJaw5-gCKrZnHLFg6d5qm-Ls6hT8gwRbIT71AA4u-oj2LwS-KXGug42Ngfld8uRtLkY3kIL3DPcmkYYJYULSpoZmxuLapKTRcgMwZEK9Ltban7dyYbfwY7UFxQSSzZ2xJ2TxYINe8T07CHJglbe7I4pWNf0eZLyKQnYawxXhBXT0lMM60yO81-q1MxFDPOQd3FN404O6w2rItM7gPy2A_R6H3iNR0IJxPb4Gm5gaxlviri9DkU3KwrHNZHXNSumrNEmNZ5-8qJhk90IQo-9OZViELEyS6OSdfKiOddj144jKwEzrV89V4aOHWqIjoHEMbqb0aFJ5BPWRU-cu9uiURwlHqFkKERDO-blpDu2YwzeTU6rM1KIMTgOiKsxF8Z6QDi5v6zm42pH3J3l.fAL6-85YCNcL6Wb4YwYnqnIBW-rrE1uvjkAkacXolII&dib_tag=se&keywords=m2%2Bheat%2Bset%2Binsert&qid=1741369080&sprefix=m2%2Bheat%2Caps%2C130&sr=8-4&th=1			

Material Choice: 3D printing ABS

ABS is the preferred choice for a weatherproof electronics enclosure due to its superior impact resistance, machinability, weather resistance, and thermal stability (for installing heat inserts). While PVC has excellent chemical resistance, it is less durable in outdoor conditions, making it less ideal for bike computer casings. [1]

PVC → not good with oil, therefore if exposed and there are screwed the threads dissolve [1]